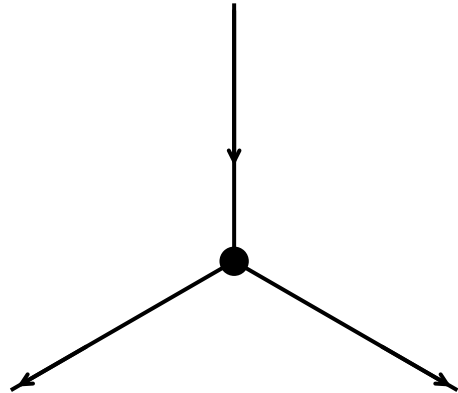


BRW interaction primitives (thesis Eqs. 3.16+3.21) via rdft.crn

V_branch (sign -1, Eq. 3.16)

$$\tilde{\psi}_a^1 \tilde{\psi}_b^0 \psi_a^2 \psi_b^0$$

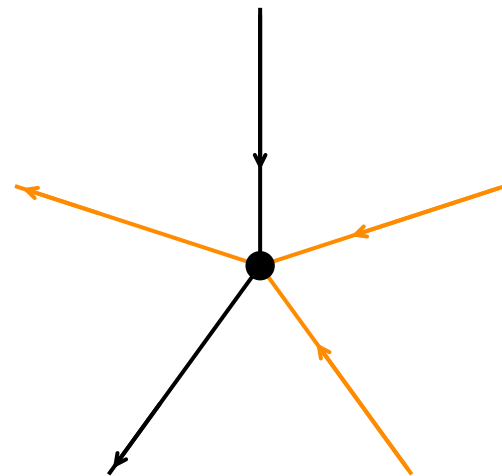
BRW branching $A \rightarrow 2A$



V_QC4 (sign -1, Eq. 3.21)

$$\tilde{\psi}_a^1 \tilde{\psi}_b^2 \psi_a^1 \psi_b^1$$

Λ -trace observable



V_QC1 (sign +1, Eq. 3.21)

$$\tilde{\psi}_a^0 \tilde{\psi}_b^1 \psi_a^1 \psi_b^0$$

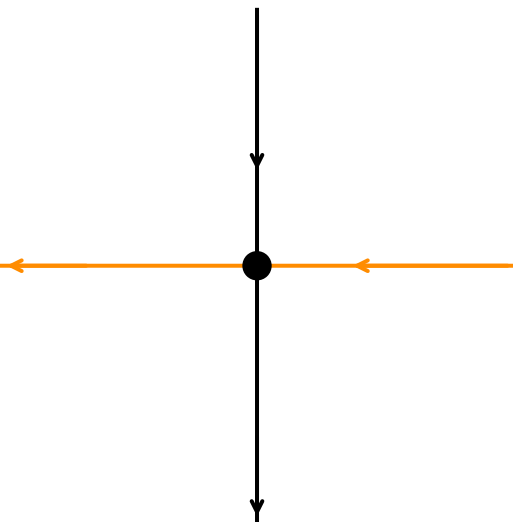
Λ -trace observable (bilinear)



V_QC5 (sign -1, Eq. 3.21)

$$\tilde{\psi}_a^1 \tilde{\psi}_b^1 \psi_a^1 \psi_b^1$$

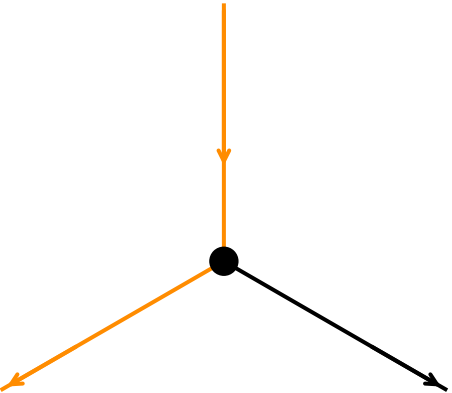
Λ -trace observable



V_QC2 (sign -1, Eq. 3.21)

$$\tilde{\psi}_a^0 \tilde{\psi}_b^1 \psi_a^1 \psi_b^1$$

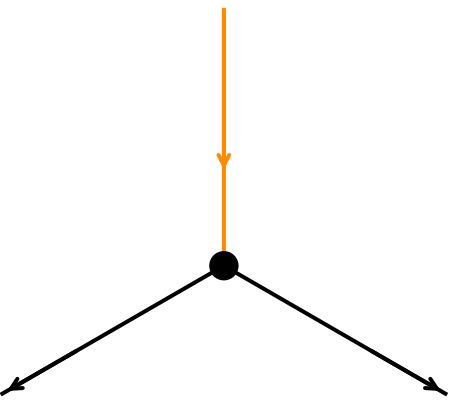
Λ -trace observable



V_QC6 (sign +1, Eq. 3.21)

$$\tilde{\psi}_a^0 \tilde{\psi}_b^1 \psi_a^2 \psi_b^0$$

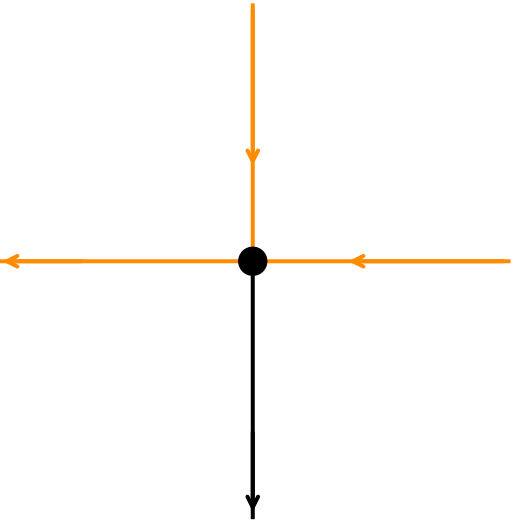
Λ -trace observable



V_QC3 (sign -1, Eq. 3.21)

$$\tilde{\psi}_a^0 \tilde{\psi}_b^2 \psi_a^1 \psi_b^1$$

Λ -trace observable



Legend

— A species ($\psi_a, \tilde{\psi}_a$)

— B species ($\psi_b, \tilde{\psi}_b$)